Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/19/2007
PI Name:	Chapes, Stephen K. Ph.D.		
Project Title:	Differentiation of Bone Marrow Macrophages in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	skcbiol@ksu.edu	**Fax:**	FY 785-532-6653
PI Organization Type:	UNIVERSITY	**Phone:**	785-532-6795
Organization Name:	Kansas State University		
PI Address 1:	1717 Claflin Rd.		
PI Address 2:	Division of Biology		
PI Web Page:			
City:	Manhattan	**State:**	KS
Zip Code:	66506-4901	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	12/28/1998	**End Date:**	03/31/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-115 NOTE: Received NCE to 3/31/2008 per A. Chu/ARC (7/07) NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1274		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space flight and/or microgravity have dramatic effects on macrophage and other cells. For example, murine bone marrow macrophages respond to microgravity in a period of about 8 seconds. These early responses to microgravity are associated with observations that bone marrow stem cell differentiation into mature macrophages is inhibited during space flight. The specific aim of this project is to determine mechanisms for the disrupted macrophage differentiation during space flight. I propose to test two hypotheses to address this specific aim: (1) Macrophage differentiation is affected because the receptor for CSF-1, c-fms is down regulated on macrophage precursors, and (2) Space flight disrupts c-fos-dependent processes involved in macrophage differentiation.We will use SHOT Inc. hardware, the Cellcult unit and bioreactor to grow primary bone marrow in the presence of macrophage colony stimulating factor. All experimental preparations will be done at the PI's home institution or at Morehouse University in Atlanta. Therefore, this project meets the requirement for needing limited crew training and involvement as described in NRA 96-HEDS-04. Moreover, this project has a flight history and a sound experimental rationale to justify its execution. Due to the changes in NASA priorities, this project was terminated and in the final grant period, our group has developed and tested tools that can be used for future shuttle missions. Indeed, some of the assays we developed over the last year will be used to assess bone marrow samples that will be obtained from a cooperative agreement with a commercial flight experiment to be done on STS-118 (Commercial Biomedical Test Module-2, CBTM-2).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight offers a unique environment that impacts many physiological systems. Amongst these systems, bone metabolism is known to be significantly altered by the skeletal unloading of space flight. Hematopoiesis is the process which provides the body with mature, functioning cells necessary for oxygenation, clotting and immune functions. The bone, specifically, the bone marrow, is the site for hematopoiesis.Moreover, changes in bone impact all compartments therein, including the marrow. Indeed, analysis of animals from rats to rhesus moneky to humans shows that hematopoiesis can be affected by space flights. Although direct examination of primary hematopoiesis is not an aim of the proposed project, this experiment will allow us to dissect a very complex system. We will obtain valuable information about what mechanisms involved in hematopoiesis are susceptible to disruption by skeletal unloading. This understanding will allow for more effective countermeasure planning. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. These prolonged periods of skeletal unloading also induce changes in bone that parallel those seen in space flight. In addition, individuals with bone disorders, such as osteoporosis, are also subject to alterations in hematopoiesis. This is best exemplified by the osteopetrotic mouse, which has diminished numbers of macrophages and brittle bones. Therefore, data obtained from this study will also have direct extrapolations to individuals on Earth.
Task Progress:	We have developed methodology and tools to assess changes in macrophages over the final funding period. We have the ability to assess the phenotypes of macrophages and determine the differentiation state of the cells. Therefore, these tools will be available to assess how space flight affects bone marrow macrophage differentiation during space flight or any other macrophage populations that can be assessed.
Bibliography Type:	Description: (Last Updated: 09/15/2021)
Articles in Peer-reviewed Journals	Ganta RR, Cheng C, Miller EC, McGuire BL, Peddireddi L, Sirigireddy KR, Chapes SK. "Differential clearance and immune responses to tick cell-derived versus macrophage culture-derived Ehrlichia chaffeensis in mice." Infect Immun. 2007 Jan;75(1):135-45. Epub 2006 Oct 23. PMID: 17060466 , Jan-2007
Papers from Meeting Proceedings	Delandre C, Penabaz TR, Passarelli AJ, Clem RJ, Chapes SK. "Functional characterization of CD81 mutants lacking intracellular and transmembrane cysteine residues. " FASEB Summer Research Conference, Tucson, AZ, July 22-27, 2006. Abstract Book, pg 58; FASEB Summer Research Conference, Tucson, AZ, July 22-27, 2006. , Jul-2006
Papers from Meeting Proceedings	Potts BE, Snyder LL, Chapes SK. "In vitro migration and tissue specific responses of C2D macrophages to adipocytes." Presented at the National IDeA Symposium of Biomedical Research Excellence (NISBRE), Washington, D.C., July 20-22, 2006. National IDeA Symposium of Biomedical Research Excellence (NISBRE), Abstracts, p. C283, 2006. , Jul-2006
Papers from Meeting Proceedings	Chapes SK, Cheng C, Miller EC, McGuire BL, Ganta RR. "Bacterial source affects the host immune response to Ehrlichia chaffeensis." Presented at the 20th Meeting of the American Society for Rickettsiology and the 5th International Conference on Bartonella as Emerging Pathogens, Pacific Grove, CA, September 2-7, 2006. Abstract Book, 20th Meeting of the American Society for Rickettsiology and the 5th International Conference on Bartonella as Emerging Pathogens, September 2006. , Sep-2006
Papers from Meeting Proceedings	McGuire BL, Cheng C, Ganta RR, Fleming RK, Chapes SK. "Distinct kinetics of humoral and cell-mediated responses to Ehrlichia chaffeensis. " Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. Journal of Leukocyte Biology 2006; S:60. , Oct-2006
Papers from Meeting Proceedings	Mordica WJ, Fleming RK, Clem RJ, Passarelli L, Chapes SK. "Creation of a CD81-/- macrophage cell line." Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. Journal of Leukocyte Biology 2006; S:85. , Oct-2006
Papers from Meeting Proceedings	Fedrow AL, Von Ohlen T, Ganta R, Chapes SK. "Drosophila S2 cells as a model to study E. chaffeensis infections." Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. Journal of Leukocyte Biology 2006;S:69. , Oct-2006

Papers from Meeting Proceedings	Potts BE, Chapes SK. "In Vivo responses and trafficking patterns of C2D macrophages after intraperitoneal adoptive transfer." Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. Journal of Leukocyte Biology 2006 S:84. , Oct-2006
Papers from Meeting Proceedings	Snyder LL, Potts BE, Chapes SK. "In vitro migration and tissue specific responses of C2D macrophages to adipocytes." Presented at the 39th Annual Meeting of the Society for Leukocyte Biology, San Antonio, TX, November 9-11, 2006. Journal of Leukocyte Biology 2006;Supplement addendum:10. , Oct-2006